

**BANARAS HINDU UNIVERSITY**

B.Voc. Semester IV Examination, 2023-24

**Paper: GEN-401 — Real Analysis**

Time: Three Hours

Maximum Marks: 70

*Instructions: Answer the questions in each section as instructed. All questions carry marks as indicated.*

**SECTION — A**

*Note: Very Short Answer Type Questions. Answer all questions. Each question carries 1 mark.* **[10 × 1 = 10]**

**Question 1.** Define the Nabla operator ( $\nabla$ ).

**Question 2.** State the Sandwich Theorem.

**Question 3.** What is a null vector?

**Question 4.** What is the dot product in  $\mathbb{R}^n$ ?

**Question 5.** Find the curl of  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ .

**Question 6.** Find the divergence of  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ .

**Question 7.** Find the gradient of  $\phi = xyz$ .

**Question 8.** Define an irrotational vector field.

**Question 9.** Define the scalar triple product of an orthogonal system.

**Question 10.** Define the uniform continuity of a real-valued function.

**SECTION — B**

*Note: Short Answer Type Questions. Answer any six questions. Each question carries 6 marks.* **[6 × 6 = 36]**

**Question 1.** State the Cauchy criterion for a series of functions.

**Question 2.** State and prove the Weierstrass M-test.

**Question 3.** Define a piecewise continuous function and give an example.

**Question 4.** Show that:  $\nabla \left( \frac{1}{r} \right) = -\frac{\vec{r}}{r^3}$ .

**Question 5.** Define the curl of a vector field. Find the curl of  $\vec{V}$  where  $\vec{V} = (xyz, xyz, xyz)$ .

**Question 6.** Define the uniform convergence of a series of functions and discuss the convergence of the series  $\sum x^n$  in  $[0, 1]$ .

**Question 7.** Define the continuity of a real-valued function. Show that  $f(x) = x$  is continuous by the  $\epsilon$ - $\delta$  definition.

**Question 8.** Show that if the function  $f$  is continuous on  $[a, b]$  and  $f$  is differentiable in  $(a, b)$  and  $f'(c) = 0$  for all  $c \in (a, b)$ , then  $f$  is constant.

**Question 9.** Determine which of the following functions are even and which are odd:

- (a)  $f(x) = \sin x$
- (b)  $f(x) = x^2 \cos x$
- (c)  $f(x) = 1$
- (d)  $f(x) = x$

## SECTION — C

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*Note: Long Answer Type Questions. Answer any two questions. Each question carries 12 marks.* **[2 × 12 = 24]**

**Question 10.**

- (a) Define the gradient of a scalar function  $\phi$  defined on  $\mathbb{R}^3$ , and also define the directional derivative of the function  $\phi$ . [6]
- (b) Find the directional derivative of  $\phi(x, y, z) = x^4y + z^2$  in the direction of  $\vec{A} = \hat{i} + \hat{j} + \hat{k}$ . [6]

**Question 11.**

- (a) Define the Fourier series. [4]
- (b) Define the square wave function and find the Fourier series of the square wave function. [8]

**Question 12.**

- (a) State and prove the Mean Value Theorem. [8]
- (b) State Rolle's Theorem. [4]

**Question 13.**

- (a) Find the unit normal vector to the ellipsoid  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$  at the point  $\left(\frac{a}{3}, \frac{b}{3}, \frac{c}{3}\right)$ . [8]
- (b) Define a periodic function. Check whether  $f(x) = \sin^2 x$  is periodic or not. [4]

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